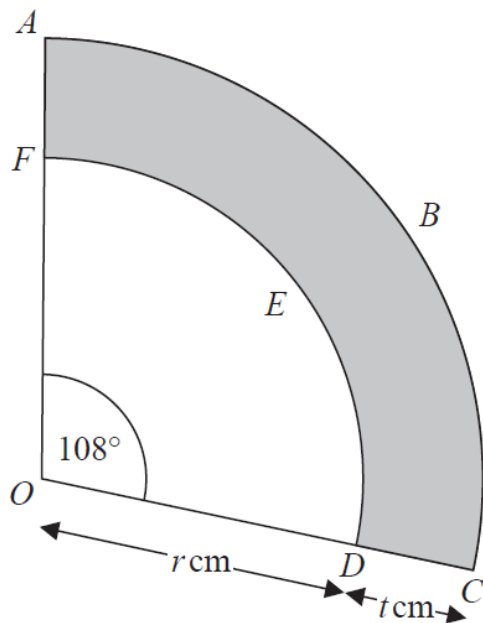




ABC and DEF are two arcs of circles, centre O .
 OFA and ODC are straight lines.

- (a) Show that the perimeter of the shaded region is given by $\frac{6\pi r + 3\pi t + 10t}{5}$ cm

(5)

(b)(i) Find the exact value of $\frac{6\pi r + 3\pi t + 10t}{5}$ when $r = 0$ and $t = 10$

Give your answer in its simplest form.

.....
(1)

(ii) Explain what your value in part (b)(i) represents.

.....
.....
.....
(1)

(Total for Question 14 is 7 marks)